An Extended Object PMB Filter with Zero-Inflated Poisson Model Using Belief Propagation

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***Abstract*—This paper presents an efficient implementation of the extended object Poisson multi-Bernoulli (PMB) filter under the zero-inflated Poisson (ZIP) object measurement model using belief propagation (BP). The ZIP measurement model separates a Bernoulli detection event from the conditional Poisson generation of object measurements, enabling principled handling of empty measurement sets. Building upon the PMB mixture posterior, we present a factorized joint posterior of set of objects with object detection variables and a dual representation of data association using both object-oriented and measurement-oriented association variables. Notably, this representation replaces the implicit high-order global hypothesis constraint by local consistency factors, yielding a factor graph amenable to BP. In addition, we present a particle-based implementation. Simulation results demonstrate that the proposed method compares well with the sampling-based extended object PMB mixture and PMB filters with ZIP model in terms of estimation accuracy and runtime.**

I. Introduction

Multi-object tracking (MOT) concerns estimating the states of multiple objects from sequences of noisy sensor measurements [1]. Modern high-resolution sensors, such as lidar and automotive radar, often produce multiple measurements from a single object. Tracking such objects is called extended object tracking (EOT), where the object state includes both kinematic and extent parameters [2]. This paper focuses on multiple EOT with an unknown and time-varying number of objects.

For EOT, the set of single object measurements is commonly modeled using an inhomogeneous Poisson point process (PPP) [3]. The zero-inflated Poisson (ZIP) model augments the PPP with an explicit Bernoulli detection event. Under this model, an object is first detected with some probability, and only if it is detected, it then generates a Poisson set of measurements. Consequently, unlike the standard PPP model, the ZIP model can account separately for object missed detections and for the Poisson variability in the number of generated measurements. This separation is particularly beneficial for multiple EOT with partially occluded objects, since it provides a more flexible and interpretable measurement model.

For standard multi-object models with Poisson birth and the general measurement model that each object generates an independent set of measurements, the exact multi-object posterior is of the Poisson multi-Bernoulli mixture (PMBM) form [4], [5]. The PMBM density admits a compact representation of global hypotheses with probabilistic object existence encoded in each Bernoulli component, and it also enables measurement-driven track initiation by representing undetected objects as a PPP [6]. The PMBM filtering recursions have been developed for point objects [6], extended objects [7] and more general measurement models [4], [5], and the Poisson multi-Bernoulli (PMB) filter is an efficient approximation of the PMBM filter that approximates the PMBM posterior as a PMB [8], [9].

A central challenge in multiple EOT is the data association problem, which must be addressed to keep the computational complexity of multiple EOT methods tractable. The most common solution is by enumerating data association hypotheses with high weights and pruning those with negligible weights. Approaches of this kind include the two-step method based on clustering and assignment (C&A) [7], [10] and sampling-based methods [11]. However, the performance of C&A-based methods often sharply drop in scenarios with objects moving in proximity, and sampling-based methods often have a high computational cost.

A more scalable approach that avoids explicit enumeration of data associations is to directly compute the marginal multi-object posterior density, where the data association uncertainties are marginalized out. This can be achieved by performing belief propagation (BP) on a factor graph representation of the joint posterior over data association variables and multi-object states [12], [13]. In particular, [13] shows that, for the extended object PMB filter, its particle BP-based implementation outperforms both C&A-based and sampling-based implementations in a challenging scenario with objects moving in proximity.

However, the BP-based implementations in [12], [13] are restricted to the PPP measurement model and do not explicitly model the object detection probability, while the C&A-based and sampling-based implementations in [7], [11] can handle the more general ZIP measurement model. The ZIP model is more challenging for BP because, when no measurements are generated, this event may result either from a missed detection under the Bernoulli detection mechanism or from an object being detected but still producing zero Poisson measurements. This ambiguity introduces an additional coupling in the joint posterior, leading to a high-order factor in the corresponding factor graph, which is not well suited for efficient BP.

A BP-based implementation for the ZIP measurement model is proposed in [14] for the analytical gamma Gaussian inverse-Wishart (GGIW) model, but it does not consider the possibility that the Poisson detection branch of the ZIP model could still generate zero measurements. Furthermore, analytical GGIW implementations rely on clustering methods to enumerate data association hypotheses in calculating the extrinsic messages for Bernoulli components, which can degrade performance in scenarios with closely-spaced objects. This has been improved in [15], where the intractable terms involved in computing the extrinsic messages are approximately computed using importance sampling. However, we find that the factorization of the joint posterior in [15, Eq. 10] is algebraically inconsistent with

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the introduced factors[1]. An alternative factor graph formulation under the ZIP measurement model is presented in [16], and the max-sum message passing is used to compute the maximum a posteriori (MAP) data association. However, retaining only the MAP solution may result in degraded performance with high data association uncertainties. Furthermore, the corresponding factor for ZIP in the factor graph remains high-order in [16], making it unsuitable for efficient BP.

In light of the discussion, there is a need for an efficient implementation of the extended object PMB filter under the ZIP measurement model that can handle the coupling introduced by the ZIP model and also avoid explicit enumeration of data association hypotheses. In this paper, we present an efficient implementation of the extended object PMB filter under the ZIP measurement model using BP. The main contributions are:

- We present a joint posterior for the PMB update under the ZIP measurement model and its factorization. In particular, by introducing auxiliary object detection variables and a dual data association representation using both object-oriented and measurement-oriented association variables, we obtain a factor graph amenable to efficient BP.
- Simulation results show that the proposed extended object PMB filter using particle BP compares well with existing sampling-based implementations [11] under the ZIP model in terms of both estimation accuracy and runtime.

II. Background

A. Multi-Object Models

The single object state $x_k \in \mathbb{R}^{d_x}$ at time step k contains information of the object, including e.g., its kinematic state and its extent state, which describes its shape and size. The set of object states at time step k is $\mathbf{x}_k = \{x_k^1, \ldots, x_k^{n_k}\} \in \mathcal{F}(\mathbb{R}^{d_x})$, and the set of measurements at time step k is $\mathbf{z}_k = \{z_k^1, \ldots, z_k^{m_k}\} \in \mathcal{F}(\mathbb{R}^{d_z})$.

1) Dynamic model: At time step k, new objects are born following a PPP with birth intensity $\lambda_k^B(x_k)$, independently of any existing objects. Each existing object x_{k-1} survives with probability $p_k^S(x_{k-1})$, and if it survives, it evolves according to the Markovian transition density $g_k(x_k|x_{k-1})$, independently of any other objects.

2) Measurement model: The set $\mathbf{z}_k$ of measurements is the union of object-generated measurements and clutter measurements. The set of clutter measurements is a PPP with Poisson intensity $\lambda_k^C(z_k) = \gamma_k^C \mu_k^C(z_k)$, where γ_k^C is the Poisson rate and $\mu_k^C(z_k)$ is the clutter spatial distribution.

A single object with state x_k generates an independent set $\mathbf{w}_k$ of measurements with density $\ell_k(\mathbf{w}_k|x_k)$. In this work, we assume that $\mathbf{w}_k$ is a ZIP process, and its set density is

$$\ell_k(\mathbf{w}_k|x_k) = \begin{cases} p_k^D(x_k) e^{-\gamma_k(x_k)} [\gamma_k(x_k)\ell_k(\cdot|x_k)]^{\mathbf{w}_k} & \mathbf{w}_k \neq \emptyset \\ 1 - p_k^D(x_k) + p_k^D(x_k) e^{-\gamma_k(x_k)} & \mathbf{w}_k = \emptyset, \end{cases} \tag{1}$$

where $p_k^D(x_k)$ denotes the detection probability. If object x_k is detected, it generates a Poisson set of measurements with rate $\gamma_k(x_k)$, and $\ell_k(\cdot|x_k)$ is the single measurement likelihood. The probability that object x_k generates at least one measurement is given by $1 - \ell_k(\emptyset|x_k) = p_k^D(x_k)(1 - e^{-\gamma_k(x_k)})$.

[1]Specifically, in the appendix of [15], Equations (B.19) and (B.22) do not reproduce the legacy-object no-association branch of Equation (B.18).

B. PMBM Density

Given a sequence of measurements $\mathbf{z}_{1:k'}$ up to and including time step $k' \in \{k-1, k\}$ and the multi-object models described in Section II-A, the exact multi-object density at time step k is of the PMBM form with [7]

$$f_{k|k'}(\mathbf{x}_k) = \textstyle\sum_{\mathbf{x}_k^u \uplus \mathbf{x}_k^d = \mathbf{x}_k} f_{k|k'}^p(\mathbf{x}_k^u) f_{k|k'}^{mbm}(\mathbf{x}_k^d), \tag{2}$$

$$f_{k|k'}^p(\mathbf{x}_k^u) = e^{-\langle \lambda_{k|k'}, 1\rangle} [\lambda_{k|k'}(\cdot)]^{\mathbf{x}_k^u}, \tag{3}$$

$$f_{k|k'}^{mbm}(\mathbf{x}_k^d) = \textstyle\sum_{a \in \mathcal{A}_{k|k'}} w_{k|k'}^a \sum_{\uplus_{l=1}^{n_{k|k'}} \mathbf{x}_k^l = \mathbf{x}_k^d} \prod_{i=1}^{n_{k|k'}} f_{k|k'}^{i,a^i}(\mathbf{x}_k^i), \tag{4}$$

where the PPP $f_{k|k'}^p(\cdot)$ has intensity $\lambda_{k|k'}(\cdot)$, and it represents undetected objects that are hypothesized to exist but have not been detected, whereas the MBM $f_{k|k'}^{mbm}(\cdot)$ represents potential objects that have been detected at least once at some point up to time step k'.

The MBM in (4) has $n_{k|k'}$ Bernoulli components. For the i-th Bernoulli component, there are $h_{k|k'}^i$ local hypotheses, and its density under local hypothesis $a^i \in \{1, \ldots, h_{k|k'}^i\}$ is

$$f_{k|k'}^{i,a^i}(\mathbf{x}_k^i) = \begin{cases} 1 - r_{k|k'}^{i,a^i} & \mathbf{x}_k^i = \emptyset \\ r_{k|k'}^{i,a^i} f_{k|k'}^{i,a^i}(x) & \mathbf{x}_k^i = \{x\} \\ 0 & \text{otherwise}, \end{cases} \tag{5}$$

where $r_{k|k'}^{i,a^i}$ is the probability of existence and $f_{k|k'}^{i,a^i}(\cdot)$ is the single object density conditioned on existence. The set $\mathcal{A}_{k|k'}$ contains all the global hypotheses [7]. Each MB component in (4) represents the density of the set $\mathbf{x}_k^d$ of detected objects conditioned on a global hypothesis $a = (a^1, \ldots, a^{n_{k|k'}}) \in \mathcal{A}_{k|k'}$, which selects a local hypothesis from each Bernoulli component. The weight of global hypothesis a satisfies $w_{k|k'}^a \propto \prod_{i=1}^{n_{k|k'}} w_{k|k'}^{i,a^i}$, where $w_{k|k'}^{i,a^i}$ is the weight of local hypothesis a^i for the i-th Bernoulli component, and normalization is required to ensure that $\sum_{a \in \mathcal{A}_{k|k'}} w_{k|k'}^a = 1$.

C. Poisson Multi-Bernoulli Approximation

The PMB is a special case of PMBM with a single MB. By augmenting the object space with auxiliary variables, the best PMB approximation can be obtained by minimizing the Kullback-Leibler divergence (KLD) [17].

To do so, we first augment the single object space with an auxiliary variable $u \in \mathbb{U}_{k|k'} = \{0, 1, \ldots, n_{k|k'}\}$, such that $(u, x) \in \mathbb{U}_{k|k'} \times \mathbb{R}^{d_x}$. A set of objects with auxiliary variables is denoted as $\widetilde{\mathbf{x}}_k \in \mathcal{F}(\mathbb{U}_{k|k'} \times \mathbb{R}^{d_x})$. Given $f_{k|k'}(\cdot)$ of the form (2), the density $f_{k|k'}(\cdot)$ on the space $\mathcal{F}(\mathbb{U}_{k|k'} \times \mathbb{R}^{d_x})$ is [17]

$$f_{k|k'}(\widetilde{\mathbf{x}}_k) = f_{k|k'}^p(\widetilde{\mathbf{y}}_k) \textstyle\sum_{a \in \mathcal{A}_{k|k'}} w_{k|k'}^a \prod_{i=1}^{n_{k|k'}} f_{k|k'}^{i,a^i}(\widetilde{\mathbf{x}}_k^i) \tag{6}$$

where, for a given $\widetilde{\mathbf{x}}_k$, $\widetilde{\mathbf{y}}_k = \{(u, x) \in \widetilde{\mathbf{x}}_k : u = 0\}$, $\widetilde{\mathbf{x}}_k^i = \{(u, x) \in \widetilde{\mathbf{x}}_k : u = i\}$, and

$$f_{k|k'}^p(\widetilde{\mathbf{y}}_k) = e^{-\langle \lambda_{k|k'}, 1\rangle} [\lambda_{k|k'}(\cdot)]^{\widetilde{\mathbf{y}}_k}, \tag{7a}$$

$$\lambda_{k|k'}(u,x) = \delta_0[u]\lambda_{k|k'}(x), \tag{7b}$$

$$f^{i,a^i}_{k|k'}(\widetilde{\mathbf{x}}^i_k) = \begin{cases} 1 - r^{i,a^i}_{k|k'} & \widetilde{\mathbf{x}}^i_k = \emptyset \\ \delta_i[u] r^{i,a^i}_{k|k'} f^{i,a^i}_{k|k'}(x) & \widetilde{\mathbf{x}}^i_k = \{(u,x)\} \\ 0 & \text{otherwise.} \end{cases} \tag{7c}$$

We notice that the sum over sets in (2) disappears in (6) due to the use of auxiliary variables since now there is only one possible partition of $\widetilde{\mathbf{x}}_k$ into $\widetilde{\mathbf{y}}_k$, $\widetilde{\mathbf{x}}^1_k, \ldots, \widetilde{\mathbf{x}}^{n_{k|k'}}_k$ that provides a non-zero density.

Given a PMBM density $f_{k|k'}(\cdot)$ of the form (6), the PMB density that minimizes the KLD from $f_{k|k'}(\cdot)$ has closed form and is given by [17, Prop. 2]

$$f^{pmb}_{k|k'}(\widetilde{\mathbf{x}}_k) = f^p_{k|k'}(\widetilde{\mathbf{y}}_k) \prod_{i=1}^{n_{k|k'}} f^i_{k|k'}(\widetilde{\mathbf{x}}^i_k), \tag{8a}$$

$$f^i_{k|k'}(\widetilde{\mathbf{x}}) = \begin{cases} 1 - r^i_{k|k'} & \widetilde{\mathbf{x}} = \emptyset \\ \delta_i[u] r^i_{k|k'} f^i_{k|k'}(x) & \widetilde{\mathbf{x}} = \{(u,x)\} \\ 0 & \text{otherwise,} \end{cases} \tag{8b}$$

$$r^i_{k|k'} = \textstyle\sum_{a^i=1}^{h^i_{k|k'}} \overline{w}^{i,a^i}_{k|k'} r^{i,a^i}_{k|k'}, \tag{8c}$$

$$f^i_{k|k'}(x) = \textstyle\sum_{a^i=1}^{h^i_{k|k'}} \overline{w}^{i,a^i}_{k|k'} r^{i,a^i}_{k|k'} f^{i,a^i}_{k|k'}(x)/r^i_{k|k'}, \tag{8d}$$

$$\overline{w}^{i,a^i}_{k|k'} = \textstyle\sum_{b \in \mathcal{A}_{k|k'}: b^i = a^i} w^b_{k|k'}. \tag{8e}$$

If we integrate out the auxiliary variables in (8), we obtain the PMB density without auxiliary variables [17]

$$f^{pmb}_{k|k'}(\mathbf{x}_k) = \textstyle\sum_{\uplus_{l=1}^{n_{k|k'}} \mathbf{x}^l \uplus \mathbf{y} = \mathbf{x}_k} f^p_{k|k'}(\mathbf{y}) \prod_{i=1}^{n_{k|k'}} f^i_{k|k'}(\mathbf{x}^i) \tag{9}$$

where the i-th Bernoulli component $f^i_{k|k'}(\mathbf{x}^i)$ is parameterized by $r^i_{k|k'}$ in (8c) and $f^i_{k|k'}(x)$ in (8d).

D. Poisson Multi-Bernoulli Update

We focus on PMB update, and details of the PMB prediction equations can be found in [9]. We first define the set of global hypotheses for extended objects. We refer to measurement $\mathbf{z}^j_k$ using the pair (k,j), and the set of all such pairs at time step k is denoted $\mathcal{M}_k$. We also let $\mathcal{M}^{i,a^i}_k \subseteq \mathcal{M}_k$ be the set of index pairs associated to local hypothesis a^i of the i-th Bernoulli component. Then, the set of all feasible global hypotheses is

$$\mathcal{A}_{k|k} = \left\{ (a^1, \ldots, a^{n_{k|k}}) : a^i \in \{1, \ldots, h^i_{k|k}\} \; \forall \; i, \biguplus_{i=1}^{n_{k|k}} \mathcal{M}^{i,a^i}_k = \mathcal{M}_k \right\}, \tag{10}$$

where the constructions of $\mathcal{M}^{i,a^i}_k$ will be given in Lemma 1. We consider the PMB update, where each measurement creates a new Bernoulli component [9], instead of each non-empty subset of measurements creating a new Bernoulli component, as in [7].

Lemma 1: Given the PMB predicted density at time step k of the form (9) and measurements $\mathbf{z}_k = \{z^1_k, \ldots, z^{m_k}_k\}$, the updated density is a PMBM of the form (2), with

$$n_{k|k} = n_{k|k-1} + m_k, \tag{11}$$

$$\mathcal{M}_k = \{(k,j) \mid j \in \{1, \ldots, m_k\}\}, \tag{12}$$

$$\lambda_{k|k}(x) = \ell_k(\emptyset|x)\lambda_{k|k-1}(x). \tag{13}$$

The i-th existing Bernoulli component, $i \in \{1, \ldots, n_{k|k-1}\}$, has $h^i_{k|k} = 2^{m_k}$ local hypotheses, either corresponding to a misdetection or an update using a non-empty subset of $\mathbf{z}_k$. For a misdetection hypothesis, we have $\mathcal{M}^{i,1}_k = \emptyset$, and

$$\ell^{i,1,0}_{k|k} = \langle f^i_{k|k-1}, \ell_k(\emptyset|\cdot)\rangle, \tag{14a}$$

$$w^{i,1}_{k|k} = 1 - r^i_{k|k-1} + r^i_{k|k-1}\ell^{i,1,0}_{k|k}, \tag{14b}$$

$$r^{i,1}_{k|k} = \frac{r^i_{k|k-1}\ell^{i,1,0}_{k|k}}{1 - r^i_{k|k-1} + r^i_{k|k-1}\ell^{i,1,0}_{k|k}}, \tag{14c}$$

$$f^{i,1}_{k|k}(x) = \ell_k(\emptyset|x) f^i_{k|k-1}(x)/\ell^{i,1,0}_{k|k}. \tag{14d}$$

Let $\mathbf{w}^1_k, \ldots, \mathbf{w}^{2^{m_k}-1}_k$ be the non-empty subsets of $\mathbf{z}_k$. For the i-th Bernoulli component $i \in \{1, \ldots, n_{k|k-1}\}$, the local hypothesis generated by a set $\mathbf{w}^j_k$ of measurements has index $a^i = 1 + j$ with $j \in \{1, \ldots, 2^{m_k} - 1\}$, and

$$\mathcal{M}^{i,a^i}_k = \{(k,p) : z^p_k \in \mathbf{w}^j_k\}, \tag{15a}$$

$$\ell^{i,a^i,j}_{k|k} = \langle f^i_{k|k-1}, \ell_k(\mathbf{w}^j_k|\cdot)\rangle, \tag{15b}$$

$$w^{i,a^i}_{k|k} = r^i_{k|k-1}\ell^{i,a^i,j}_{k|k}, \tag{15c}$$

$$r^{i,a^i}_{k|k} = 1, \tag{15d}$$

$$f^{i,a^i}_{k|k}(x) = \ell_k(\mathbf{w}^j_k|x) f^i_{k|k-1}(x)/\ell^{i,a^i,j}_{k|k}. \tag{15e}$$

Each new Bernoulli component has a different number of local hypotheses, each created by a subset of $\mathbf{z}_k$. The set of subsets of measurements associated to the i-th new Bernoulli component ($i \in \{1, \ldots, m_k\}$) can be recursively built as [9]

$$\mathcal{S}_i = \{\{z^i_k\}\} \cup (\cup_{\mathbf{w} \in \cup_{j=1}^{i-1} \mathcal{S}_j} \{\{z^i_k\} \cup \mathbf{w}\}), \tag{16}$$

with $\mathcal{S}_1 = \{\{z^1_k\}\}$. This implies that the i-th new Bernoulli component must contain the local hypothesis created by the single measurement z^i_k and cannot be associated with measurements with index larger than i. Further, we let $\mathbf{w}^{i,\iota}_k \in \mathcal{S}_i$ denote the ι-th non-empty subset of measurements associated to the i-th new Bernoulli component, with $i = n_{k|k-1} + j$, $j \in \{1, \ldots, m_k\}$, $\iota \in \{1, \ldots, 2^{j-1}\}$. The i-th new Bernoulli component has $h^i_{k|k} = 2^{j-1} + 1$ local hypotheses. The case corresponding to non-existence is given by

$$\mathcal{M}^{i,1}_k = \emptyset, \quad w^{i,1}_{k|k} = 1, \quad r^{i,1}_{k|k} = 0, \tag{17}$$

and the others ($a^i = \iota + 1, \iota \in \{1, \ldots, 2^{j-1}\}$), corresponding to detection with $\mathbf{w}^{i,\iota}_k$, are given by

$$\mathcal{M}^{i,a^i}_k = \{(k,p) : z^p_k \in \mathbf{w}^{i,\iota}_k\}, \tag{18a}$$

$$\ell^{i,\iota}_{k|k} = \langle \lambda_{k|k-1}, \ell_k(\mathbf{w}^{i,\iota}_k|\cdot)\rangle, \tag{18b}$$

$$w^{i,a^i}_{k|k} = \delta_1[|\mathbf{w}^{i,\iota}_k|]\lambda^C_k(z^j_k) + \ell^{i,\iota}_{k|k}, \tag{18c}$$

$$r^{i,a^i}_{k|k} = \ell^{i,\iota}_{k|k}/w^{i,a^i}_{k|k}, \tag{18d}$$

$$f^{i,a^i}_{k|k}(x) = \ell_k(\mathbf{w}^{i,\iota}_k|x)\lambda_{k|k-1}(x)/\ell^{i,\iota}_{k|k}. \tag{18e}$$

III. Joint Posterior and its Factorization

A. Joint Posterior of Set of Objects and Global Hypothesis

We define the joint posterior of the set of objects $\widetilde{\mathbf{x}}_k$ and global hypothesis a on the space $\mathcal{F}(\mathbb{U}_{k|k} \times \mathbb{R}^{d_x}) \times \mathcal{A}_{k|k}$ as

$$\begin{aligned} f_{k|k}(\widetilde{\mathbf{x}}_k, a) &= f^p_{k|k}(\widetilde{\mathbf{x}}^u_k) w^a_{k|k} \textstyle\prod_{i=1}^{n_{k|k}} f^{i,a^i}_{k|k}(\widetilde{\mathbf{x}}^i_k), \\ &\propto f^p_{k|k}(\widetilde{\mathbf{x}}^u_k) \textstyle\prod_{i=1}^{n_{k|k}} w^{i,a^i}_{k|k} f^{i,a^i}_{k|k}(\widetilde{\mathbf{x}}^i_k), \\ &\triangleq f^p_{k|k}(\widetilde{\mathbf{x}}^u_k) \textstyle\prod_{i=1}^{n_{k|k}} g^{i,a^i}_{k|k}(\widetilde{\mathbf{x}}^i_k), \end{aligned} \tag{19}$$

where $g^{i,a^i}_{k|k}(\cdot)$ correspond to unnormalized Bernoulli densities and can be computed as follows according to Lemma 1.

Specifically, we have that, for missed detection hypotheses, with $i \in \{1, \ldots, n_{k|k-1}\}$,

$$g^{i,1}_{k|k}(\widetilde{\mathbf{x}}^i_k) = \begin{cases} \delta_i[u] r^i_{k|k-1} \ell_k(\emptyset|x) f^i_{k|k-1}(x) & \widetilde{\mathbf{x}}^i_k = \{(u,x)\} \\ 1 - r^i_{k|k-1} & \widetilde{\mathbf{x}}^i_k = \emptyset \\ 0 & \text{otherwise,} \end{cases} \tag{20}$$

and for detection hypotheses, with $a^i = 1 + j$, where $j \in \{1, \ldots, 2^{m_k} - 1\}$,

$$g^{i,a^i}_{k|k}(\widetilde{\mathbf{x}}^i_k) = \begin{cases} \delta_i[u] r^i_{k|k-1} \ell_k(\mathbf{w}^j_k|x) f^i_{k|k-1}(x) & \widetilde{\mathbf{x}}^i_k = \{(u,x)\} \\ 0 & \text{otherwise.} \end{cases} \tag{21}$$

For the non-existence hypotheses for new Bernoulli components, with $i = n_{k|k-1} + j, j \in \{1, \ldots, m_k\}$,

$$g^{i,1}_{k|k}(\widetilde{\mathbf{x}}^i_k) = \begin{cases} 1 & \widetilde{\mathbf{x}}^i_k = \emptyset \\ 0 & \text{otherwise,} \end{cases} \tag{22}$$

and for the existence hypotheses for new Bernoulli components, with $a^i = \iota + 1$, $\iota \in \{1, \ldots, 2^{j-1}\}$

$$g^{i,a^i}_{k|k}(\widetilde{\mathbf{x}}^i_k) = \begin{cases} \delta_i[u] \ell_k(\mathbf{w}^{i,\iota}_k|x) \lambda_{k|k-1}(x) & \widetilde{\mathbf{x}}^i_k = \{(u,x)\} \\ \delta_1[|\mathbf{w}^{i,\iota}_k|] \lambda^C_k(z^j_k) & \widetilde{\mathbf{x}}^i_k = \emptyset \\ 0 & \text{otherwise.} \end{cases} \tag{23}$$

It can be seen that the joint posterior of the set of objects and the global hypothesis in (19) already admits a factorized form. Nevertheless, this form is not directly amenable to BP, since the global hypothesis $a \in \mathcal{A}_{k|k}$ is subject to the constraint in (10), which acts as an implicit high-order factor that couples the local hypotheses globally, thereby preventing a convenient decomposition into low-order factors for efficient BP.

B. Joint Posterior Factorization with Auxiliary Variables

To make the posterior more amenable to message passing, we first introduce binary object detection variables and dual representations of object-oriented and measurement-oriented data association variables in order to expose the latent structure of the PMB update with ZIP measurements. Specifically, for the i-th existing Bernoulli component, $i \in \{1, \ldots, n_{k|k-1}\}$, we first introduce a binary detection variable $D^i_k \in \{0, 1\}$, where $D^i_k = 1$ indicates that the object is detected at time step k (and generates a Poisson number $\gamma_k(\cdot)$ of measurements which can still be zero) while $D^i_k = 0$ indicates a missed detection that corresponds to the Bernoulli non-detection branch. Note that we do not need to introduce D^i_k for new Bernoulli components that represent newly detected objects. Then, we have

$$\ell^D_k(\mathbf{w}_k, D^i_k|\widetilde{\mathbf{x}}^i_k) = \ell_k(\mathbf{w}_k|\widetilde{\mathbf{x}}^i_k, D^i_k) f^{D,i}_k(D^i_k|\widetilde{\mathbf{x}}^i_k), \tag{24}$$

where

$$f^{D,i}_k(D^i_k|\widetilde{\mathbf{x}}^i_k) = \begin{cases} \delta_i[u] p^D_k(x) e^{-\gamma_k(x)} & \widetilde{\mathbf{x}}^i_k = \{(u,x)\}, D^i_k = 1 \\ \delta_i[u](1 - p^D_k(x)) & \widetilde{\mathbf{x}}^i_k = \{(u,x)\}, D^i_k = 0 \\ 1 & \widetilde{\mathbf{x}}^i_k = \emptyset, D^i_k = 0 \\ 0 & \text{otherwise,} \end{cases} \tag{25}$$

$$\begin{aligned} &\ell_k(\mathbf{w}_k|\widetilde{\mathbf{x}}^i_k = \{(u,x)\}, D^i_k) \\ &= \begin{cases} \delta_i[u][\gamma_k(x)\ell_k(\cdot|x)]^{\mathbf{w}_k} & D^i_k = 1 \\ \delta_i[u] & D^i_k = 0, \mathbf{w}_k = \emptyset \\ 0 & \text{otherwise,} \end{cases} \end{aligned} \tag{26}$$

$$\ell_k(\mathbf{w}_k|\widetilde{\mathbf{x}}^i_k = \emptyset, D^i_k) = \begin{cases} 1 & D^i_k = 0, \mathbf{w}_k = \emptyset \\ 0 & \text{otherwise.} \end{cases} \tag{27}$$

If we marginalize D^i_k from (24), we recover the original ZIP measurement set density. By separating the Bernoulli object detection event from the conditional Poisson measurements, we obtain a more explicit factorized representation (24).

We proceed to describe how to decompose the high-order factor induced by the global hypothesis constraint $a \in \mathcal{A}_{k|k}$ into a collection of lower-order factors suitable for BP. To do so, we introduce a dual representation of the data association, one object-oriented and one measurement-oriented. Although redundant, both representations are equivalent in the sense that each admissible realization of either representation under the imposed constraints uniquely determines a global hypothesis. This redundancy allows the original global hypothesis constraint in (10) to be replaced by local consistency factors.

We first introduce the object-oriented association variables $\alpha_k = [(\alpha^1_k)^T, \ldots, (\alpha^{n_{k|k}}_k)^T]^T$, where

$$\alpha^i_k = \begin{cases} [\alpha^{i,1}_k, \ldots, \alpha^{i,m_k}_k]^T & i \in \{1, \ldots, n_{k|k-1}\}, \\ [\alpha^{i,1}_k, \ldots, \alpha^{i,j}_k]^T & i = n_{k|k-1} + j, \\ & j \in \{1, \ldots, m_k\}, \end{cases} \tag{28}$$

with $\alpha^{i,j}_k \in \{0, 1\}$, and $\alpha^{i,j}_k = 1$ if and only if the j-th measurement z^j_k is associated with the i-th Bernoulli component. Note that the object-oriented association vector corresponding to the j-th new Bernoulli component has length j. This follows directly from the local hypothesis structure of new Bernoulli components described in Lemma 1.

In addition, according to the conditional measurement set density (24), if the i-th Bernoulli component is not detected, then no measurement can be associated with it. To encode this relation, we introduce the binary factor

$$\Phi^{i,j}_k(D^i_k, \alpha^{i,j}_k) = \begin{cases} 1 & D^i_k = 1 \text{ or } \alpha^{i,j}_k = 0 \\ 0 & \text{otherwise,} \end{cases} \tag{29}$$

for $i \in \{1, \ldots, n_{k|k-1}\}$ and $j \in \{1, \ldots, m_k\}$. The product of these factors over j ensures that no measurement can be associated with the i-th Bernoulli component when $D^i_k = 0$,

$$f_{k|k}(\widetilde{\mathbf{x}}_k, D_k^{1:n_{k|k-1}}, \alpha_k, \beta_k) \propto f_{k|k}^{p}(\widetilde{\mathbf{x}}_k^u) \prod_{i=1}^{n_{k|k-1}} [f_k^{D,i}(D_k^i|\widetilde{\mathbf{x}}_k^i) \underline{f}_{k|k-1}^{i}(\widetilde{\mathbf{x}}_k^i) \prod_{j=1}^{m_k} \Phi_k^{i,j}(D_k^i, \alpha_k^{i,j})] \prod_{i=n_{k|k-1}+1}^{n_{k|k}} \overline{f}_{k|k-1}^{i}(\widetilde{\mathbf{x}}_k^i)$$
$$\times \prod_{j=1}^{m_k} [\bar{s}_k^{n_{k|k-1}+j}(\widetilde{\mathbf{x}}_k^{n_{k|k-1}+j}, \alpha_k^{n_{k|k-1}+j,j}; z_k^j) \prod_{\substack{i\in\{1,\ldots,n_{k|k-1}\}\cup \\ \{n_{k|k-1}+j+1,\ldots,n_{k|k}\}}} \underline{s}_k^i(\widetilde{\mathbf{x}}_k^i, \alpha_k^{i,j}; z_k^j) \prod_{\substack{i\in\{1,\ldots,n_{k|k-1}\}\cup \\ \{n_{k|k-1}+j,\ldots,n_{k|k}\}}} \Psi_k^{i,j}(\alpha_k^{i,j}, \beta_k^j)]. \tag{30}$$

while the case with $D_k^i = 1$ and all-zero $\alpha_k^{i,j}$ is still allowed, which represents that the corresponding object is detected but generates zero measurements.

We can rewrite the unnormalized Bernoulli densities $g_{k|k}^{i,a^i}(\cdot)$ using the object-oriented association variables α_k and binary detection variables $D_k^{1:n_{k|k-1}} = [D_k^1, \ldots, D_k^{n_{k|k-1}}]^T$, leveraging the factorization of the conditional Poisson measurement set density (24). Specifically, for the i-th Bernoulli component with $i \in \{1, \ldots, n_{k|k-1}\}$, we have

$$g_{k|k}^{D,i,\alpha_k^i}(\widetilde{\mathbf{x}}_k^i, D_k^i) = f_k^{D,i}(D_k^i|\widetilde{\mathbf{x}}_k^i) \underline{f}_{k|k-1}^{i}(\widetilde{\mathbf{x}}_k^i) \times \prod_{j=1}^{m_k} \Phi_k^{i,j}(D_k^i, \alpha_k^{i,j}) \underline{s}_k^i(\widetilde{\mathbf{x}}_k^i, \alpha_k^{i,j}; z_k^j), \tag{31}$$

and for $i = n_{k|k-1} + j$, $j \in \{1, \ldots, m_k\}$,

$$g_{k|k}^{i,\alpha_k^i}(\widetilde{\mathbf{x}}_k^i) = \overline{f}_{k|k-1}^{i}(\widetilde{\mathbf{x}}_k^i) \bar{s}_k^i(\widetilde{\mathbf{x}}_k^i, \alpha_k^{i,j}; z_k^j) \times \prod_{l=1}^{j-1} \underline{s}_k^i(\widetilde{\mathbf{x}}_k^i, \alpha_k^{i,l}; z_k^l). \tag{32}$$

In (31) and (32), the local hypothesis a^i of the i-th Bernoulli component is now represented explicitly by the corresponding object-oriented association variable α_k^i (28), which provides an equivalent description of the measurement subset associated with that component. In addition, the new factors are

$$\underline{f}_{k|k-1}^{i}(\widetilde{\mathbf{x}}_k^i) = \begin{cases} \delta_i[u] r_{k|k-1}^i f_{k|k-1}^i(x) & \widetilde{\mathbf{x}}_k^i = \{(u,x)\} \\ 1 - r_{k|k-1}^i & \widetilde{\mathbf{x}}_k^i = \emptyset \\ 0 & \text{otherwise,} \end{cases} \tag{33}$$

$$\overline{f}_{k|k-1}^{i}(\widetilde{\mathbf{x}}_k^i) = \begin{cases} \delta_i[u] p_k^D(x) e^{-\gamma_k(x)} \lambda_{k|k-1}(x) & \widetilde{\mathbf{x}}_k^i = \{(u,x)\} \\ 1 & \widetilde{\mathbf{x}}_k^i = \emptyset \\ 0 & \text{otherwise,} \end{cases} \tag{34}$$

$$\underline{s}_k^i(\widetilde{\mathbf{x}}_k^i, \alpha_k^{i,j}; z_k^j) = \begin{cases} \delta_i[u] \gamma_k(x) \ell_k(z_k^j|x) & \widetilde{\mathbf{x}}_k^i = \{(u,x)\}, \alpha_k^{i,j} = 1 \\ 1 & \alpha_k^{i,j} = 0 \\ 0 & \text{otherwise,} \end{cases} \tag{35}$$

$$\bar{s}_k^i(\widetilde{\mathbf{x}}_k^i, \alpha_k^{i,j}; z_k^j) = \begin{cases} \delta_i[u] \gamma_k(x) \ell_k(z_k^j|x) & \widetilde{\mathbf{x}}_k^i = \{(u,x)\}, \alpha_k^{i,j} = 1 \\ \lambda_k^C(z_k^j) & \widetilde{\mathbf{x}}_k^i = \emptyset, \alpha_k^{i,j} = 1 \\ 1 & \widetilde{\mathbf{x}}_k^i = \emptyset, \alpha_k^{i,j} = 0 \\ 0 & \text{otherwise.} \end{cases} \tag{36}$$

In the above, the factors (33) and (34) represent the predicted and prior information for existing and newly detected objects, respectively, without being associated with any measurements, whereas the factors (35) and (36) represent the local likelihood of associating a particular measurement with its corresponding Bernoulli component for existing and newly detected objects, respectively.

We proceed to introduce the measurement-oriented association variables $\beta_k = [\beta_k^1, \ldots, \beta_k^{m_k}]^T$, where the j-th element $\beta_k^j = i$, $j \in \{1, \ldots, m_k\}$, if and only if the j-th measurement z_k^j is associated with the i-th Bernoulli component. According to the local hypothesis structure for new Bernoulli components described in Lemma 1, the j-th measurement z_k^j cannot be associated with a new Bernoulli component, whose index is smaller than $n_{k|k-1} + j$. Therefore, for each $j \in \{1, \ldots, m_k\}$, $\beta_k^j \in \{1, \ldots, n_{k|k-1}, n_{k|k-1} + j, \ldots, n_{k|k}\}$.

To eliminate the implicit high-order factor induced by the global hypothesis constraint, we introduce binary consistency factors between the object-oriented α_k and the measurement-oriented β_k association variables. Specifically, for each valid pair (i, j), we define

$$\Psi_k^{i,j}(\alpha_k^{i,j}, \beta_k^j) = \begin{cases} 1 & \alpha_k^{i,j} = 1, \beta_k^j = i \\ 1 & \alpha_k^{i,j} = 0, \beta_k^j \neq i \\ 0 & \text{otherwise.} \end{cases} \tag{37}$$

This factor enforces that the dual representations of the data association agree locally on whether the j-th measurement z_k^j is associated with the i-th Bernoulli component. As a result, the original global hypothesis constraint in (10) is represented explicitly through a product of consistency factors between the dual association variables, thereby transforming the remaining high-order dependency into a BP-friendly local factorization.

By plugging (31) and (32) into (19) and using the consistency factors between the dual representation of data association, we can factorize the joint posterior of the set of objects $\widetilde{\mathbf{x}}_k$, object detection variables $D_k^{1:n_{k|k-1}}$, object-oriented α_k and measurement-oriented β_k association variables as in (30). The factor graph corresponding to this factorized joint posterior is illustrated in Fig. 1.

C. Loopy Belief Propagation

The objective is to compute the marginal Bernoulli density $f_{k|k}^i(\widetilde{\mathbf{x}}_k^i)$, cf. (8b), for $i \in \{1, \ldots, n_{k|k}\}$. Since the factor graph in Fig. 1 contains cycles, exact BP is no longer applicable, and we instead resort to loopy BP, with all messages updated in parallel as in [12], [13]. Due to page limits, in this section we only present the key steps of loopy BP:

1) Messages from Bernoulli state node $\widetilde{\mathbf{x}}_k^i$ to local likelihood factor node $\underline{s}_k^i$ or $\bar{s}_k^i$.
2) Messages from local likelihood factor node $\underline{s}_k^i$ or $\bar{s}_k^i$ to object-oriented association node $\alpha_k^{i,j}$.
3) Messages passing in the object detection subgraph
$$\widetilde{\mathbf{x}}_k^i - f_k^{D,i} - D_k^i - \{\Phi_k^{i,j}\}_{j=1}^{m_k} - \{\alpha_k^{i,j}\}_{j=1}^{m_k}.$$

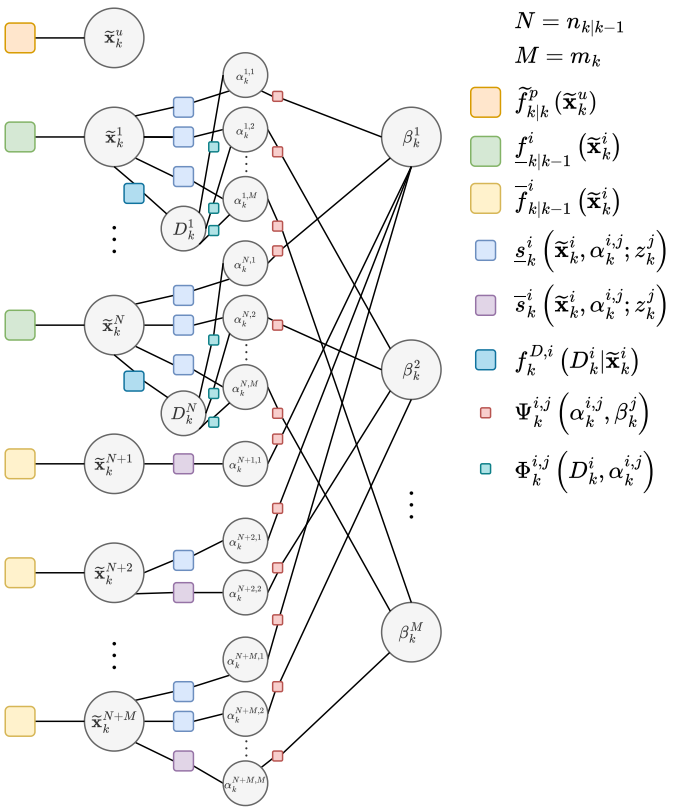


Fig. 1. Factor graph for the factorized joint posterior (30), where variable and factor nodes are represented using circles and squares, respectively.

for existing Bernoulli components.

4) Messages passing through the data association consistency factor $\Psi^{i,j}_{k}$.
5) Belief calculation for the Bernoulli state node at the last iteration.

The message passing equations under the ZIP measurement model are more intricate compared with those in [12], [13] under the Poisson measurement model due to the additional components introduced in the factor graph. We refer the reader to the extended version of this paper [18] for the detailed sum-product message passing equations and their derivation.

IV. Simulations and Results

In this section, we present the results from a Monte Carlo simulation study with 100 runs where the performance of the following four multiple extended object filters are compared:

- The proposed extended object PMB filter under the ZIP measurement model, referred to as PMB-BP-ZIP.
- The extended object PMB filter under the Poisson measurement model with BP, referred to as PMB-BP-PPP.
- The extended object PMBM filter under the ZIP measurement model with sampling-based data association [11], referred to as PMBM-SO-ZIP.
- The extended object PMB filter under the ZIP measurement model with sampling-based data association [11], referred to as PMB-SO-ZIP.

Both PMB-BP-ZIP and PMB-BP-PPP use particle implementations. Due to page limits, we kindly refer the readers to [18] and [13] for their implementation details, respectively.

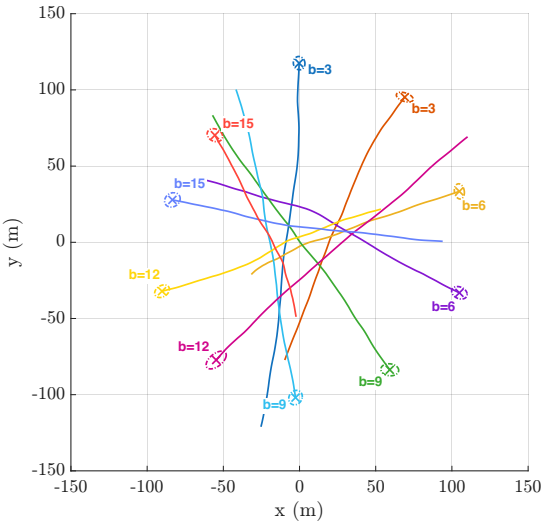


Fig. 2. Illustration of the ground-truth scenario. Two objects are born at each of the time steps 3, 6, 9, 12, and 15, while two objects are dead at time steps 83, 86, 89, 92, and 95.

The single object state is $x_k = (e_k, E_k, \gamma_k)$, consisting of the kinematic state $e_k \in \mathbb{R}^4$, a 2×2 symmetric positive definite matrix representing the extent state E_k, and the Poisson object measurement rate $\gamma_k > 0$. The kinematic state e_k consists of 2D position and velocity. We assume that each object moves according to a nearly constant velocity motion model and that its extent and measurement rate remain unchanged over time. The kinematic state transition density is

$$g_k(e_k|e_{k-1}) = \mathcal{N}(e_k; Fe_{k-1}, Q),$$
$$F = I_2 \otimes \begin{bmatrix} 1 & T \\ 0 & 1 \end{bmatrix}, \quad Q = \sigma_q^2 I_2 \otimes \begin{bmatrix} T^3/3 & T^2/2 \\ T^2/2 & T \end{bmatrix},$$

where I_2 is the 2×2 identity matrix, $T = 0.2$ is the sampling interval, and $\sigma_q = 0.8$ is the process noise standard deviation.

We consider the same scenario as in [12], [13] with a region of interest of size $[-150\,\text{m}, 150\,\text{m}] \times [-150\,\text{m}, 150\,\text{m}]$, where the ten objects start moving towards the center from positions uniformly placed on a circle of radius 125 m around the center with an initial velocity 12.5 m/s, and then they become closely-spaced for some time before they separate. The initial extent matrix of each object is obtained by drawing a sample from an inverse-Wishart distribution with degree of freedom 100 and mean matrix $5I_2$. The true object trajectories are illustrated in Fig. 2. The object survival probability is $p^S = 0.99$.

PMB-BP-ZIP and PMB-BP-PPP use particle implementations, whereas PMBM-SO-ZIP and PMB-SO-ZIP adopt analytical GGIW implementations. For all the implementations, the Poisson birth rate is 0.01. In the birth density, the velocity is Gaussian distributed with zero mean and covariance $225I_2$, the extent is inverse-Wishart distributed with mean $5I_2$ and degree of freedom 100, and the measurement rate is Gamma distributed with mean γ and rate parameter 100. The position is Gaussian distributed with zero mean and covariance $150^2 I_2$ for GGIW implementations, and uniformly distributed in the entire region for particle-based implementations. In addition, for particle-based implementations, the proposal distribution for the kinematic state is given by its motion model, whereas for the extent matrix its proposal is given by a mean-preserving

TABLE I
MULTI-OBJECT FILTERING PERFORMANCE EVALUATED USING THE GOSPA METRIC, TOGETHER WITH ITS DECOMPOSITION INTO STATE ESTIMATION, MISSED DETECTION, AND FALSE DETECTION ERRORS, AS WELL AS RUNTIME (IN SECONDS), AVERAGED OVER 100 TIME STEPS.

	$p^D = 0.9$, $\gamma = 10$					$p^D = 0.9$, $\gamma = 5$					$p^D = 0.8$, $\gamma = 10$				
	Total	State	Miss	False	Runtime	Total	State	Miss	False	Runtime	Total	State	Miss	False	Runtime
PMB-BP-ZIP	**7.39**	5.32	0.80	1.26	219.5	10.16	7.16	0.70	2.29	119.1	**8.80**	5.68	0.81	2.31	197.1
PMB-BP-PPP	63.39	10.88	6.62	45.89	374.1	13.36	7.47	3.59	2.29	118.6	62.81	12.90	12.78	37.13	325.3
PMBM-SO-ZIP	7.76	5.34	1.51	0.90	576.2	**9.76**	7.20	1.08	1.49	324.5	8.92	5.67	1.49	1.76	544.9
PMB-SO-ZIP	9.90	5.19	3.81	0.91	354.3	11.94	6.95	2.96	1.03	173.1	11.46	5.41	4.33	1.72	320.1

Wishart distribution with degree of freedom 10000, and for the measurement rate its proposal is given by a mean-preserving gamma distribution with rate parameter 10000. The details of the GGIW prediction can be found in [7].

For the measurement model, the single measurement likelihood is given by $\ell_k(z|x_k) = \mathcal{N}(z; He_k, E_k)$, $H = I_2 \otimes [1\ 0]$. The object detection probability is p^D, and the Poisson object measurement rate is γ. In the simulation, we consider three different settings with $(p^D, \gamma) \in \{(0.9, 10), (0.9, 5), (0.8, 10)\}$. The Poisson clutter measurements are uniformly distributed in the region of interest with rate $\lambda^C = 10$. As PMB-BP-PPP assumes $p^D = 1$, we set the parameter γ used in PMB-BP-PPP to be $p^D\gamma$ for a fair comparison.

For PMB-BP-ZIP and PMB-BP-PPP, the number of particles is 5000 with systematic resampling applied at every time step, and the number of iterations for loopy BP is 3. In addition, we use message censoring and measurement reordering to facilitate track initialization and also adopt measurement-driven initialization for newly detected objects [19]. Moreover, we discard Bernoulli components with existence probability smaller than 10^{-3}. For PMBM-SO-ZIP and PMB-SO-ZIP, the pruning thresholds for global hypotheses, GGIW components in the Poisson intensity and Bernoulli components are set to 10^{-2}, 10^{-3} and 10^{-3}, respectively. We also apply ellipsoidal gating with a gate size 13.8. The number of sampling iterations for data association is set to $\lceil 100 m_k w_a \rceil$ for global hypothesis with weight w_a in PMBM-SO-ZIP and $50m_k$ in PMB-SO-ZIP.

To extract object state estimates from the filtering density, for all the filters, we apply a threshold of 0.5 on the existence probabilities of Bernoulli components. As for PMBM-SO-ZIP, since it maintains multiple global hypotheses, we first select the global hypothesis with the highest weight and then extract estimates from its Bernoulli components. The performance is evaluated using the generalized optimal subpattern assignment (GOSPA) metric [20] with $\alpha = 2$, exponent $p = 1$, cut-off $c = 20$. The base measure in GOSPA is the square root Gaussian Wasserstein distance for comparing two extended object states with elliptical shapes [21].

Table I summarizes the performance of the different filters in terms of GOSPA error, its decomposition into state estimation, missed detection, and false detection errors, as well as runtime. The results show that PMB-BP-ZIP achieves the best overall performance in terms of GOSPA across all three settings. In particular, PMB-BP-ZIP significantly outperforms PMB-BP-PPP, which performs poorly due to measurement model mismatch, especially in the case of higher Poisson measurement rate. Compared to PMBM-SO-ZIP and PMB-SO-ZIP, PMB-BP-ZIP has lower GOSPA error in two out of three settings, and it has a significantly lower missed detection error than both PMBM-SO-ZIP and PMB-SO-ZIP across all three settings. In addition, PMB-BP-ZIP has a lower runtime than both PMBM-SO-ZIP and PMB-SO-ZIP.

It is worth noting that the differences in GOSPA errors between PMB-BP-ZIP and PMBM-SO-ZIP are less pronounced than those reported in [13] for a similar scenario but with the Poisson object measurement model. This may be attributed to the fact that the ZIP measurement model introduces additional loops in the factor graph, which can degrade the accuracy of the loopy BP approximation.

To further analyze the filtering performance of PMB-BP-ZIP, PMBM-SO-ZIP, and PMB-SO-ZIP, Fig. 3 presents the GOSPA error and its decomposition over time for the three settings. It can be observed that PMB-BP-ZIP yields a lower missed detection error than both PMBM-SO-ZIP and PMB-SO-ZIP across all three settings, especially at the time steps when objects are born and when closely spaced objects separate. The superior track initialization performance of PMB-BP-ZIP can be attributed to its measurement-driven proposal distribution for new Bernoulli components, although this comes at the cost of increased sensitivity to false object detections. It is also interesting to note that, although both PMB-BP-ZIP and PMB-SO-ZIP employ an MB approximation, only PMB-SO-ZIP suffers from coalescence, while PMB-BP-ZIP performs significantly better when objects are moving in proximity. This advantage stems from its particle-based representation, which can better capture posterior multi-modality than the parametric representation adopted in PMBM-SO-ZIP and PMB-SO-ZIP.

V. CONCLUSIONS

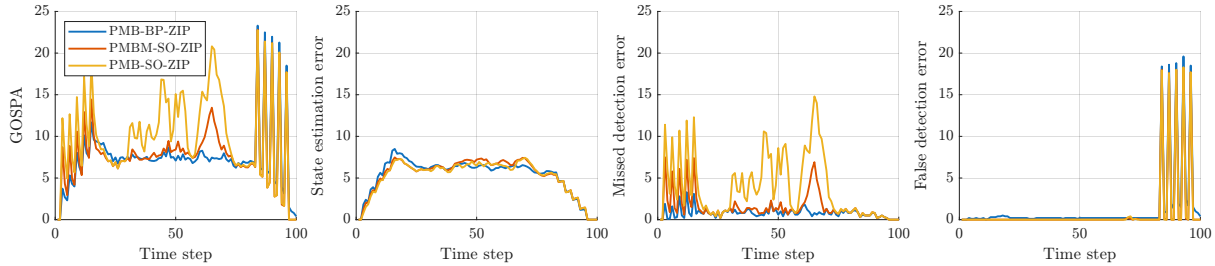
In this paper, we have presented an extended object PMB filter under the ZIP object measurement model and its implementation using particle BP. The results have shown that the proposed filter achieves better performance compared to the extended object PMBM filter using sampling-based method in terms of estimation error and runtime. Future work includes validation on real-world automotive radar and lidar data.

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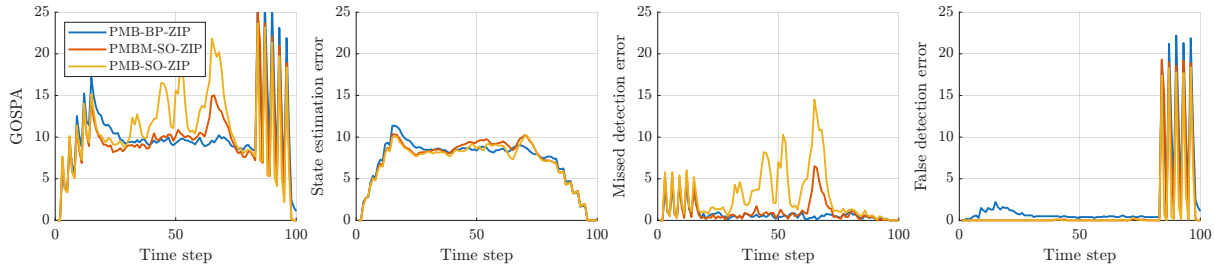
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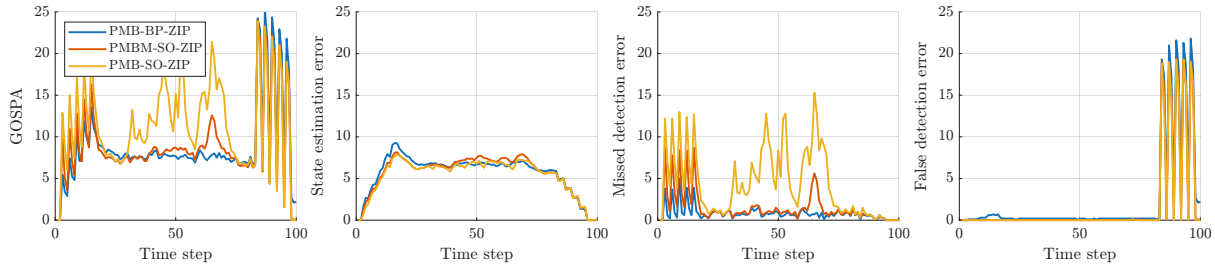
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(a) $p^D = 0.9$, $\gamma = 10$



(b) $p^D = 0.9$, $\gamma = 5$



(c) $p^D = 0.8$, $\gamma = 10$

Fig. 3. Performance comparison of PMB-BP-ZIP, PMBM-SO-ZIP and PMB-SO-ZIP, showing the GOSPA error and its decomposition over time.

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